FINGERPRINT PATTERN RECOGNITION FOR BLOOD GROUP CLASSIFICATION USING DEEP LEARNING

A PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

The work embodied in the present project report entitled "**Fingerprint Pattern Recognition For Blood Group Classification Using Deep Learning**" has been carried out by the students SAHAANA S, SHARVINI S, THENNARASI M, The work reported herein is original and we declare that the project is their own work, except where specifically acknowledged, and has not been copied from other sources or been previously submitted for assessment.

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ABSTRACT

Blood group identification is an important process in medical diagnosis, blood transfusion, and emergency healthcare services. Traditionally, blood group detection requires laboratory testing using blood samples and chemical reagents, which can be time-consuming and require medical facilities. Fingerprints are unique biometric traits that contain ridge patterns which may have correlations with certain physiological attributes, including blood groups. The project proposes a deep learning-based system for predicting blood groups using fingerprint images. The system uses Convolutional Neural Network (CNN) architectures such as VGG16, ResNet, AlexNet, and LeNet to extract features from fingerprint patterns and classify them into eight blood group categories: A+, A-, B+, B-, AB+, AB-, O+, and O-. The fingerprint images are preprocessed and used to train the deep learning models to recognize pattern variations.

Keywords: Fingerprint Recognition, Blood Group Detection, Deep Learning, Convolutional Neural Network (CNN), Image Processing, Biometric Identification, Machine Learning.

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LIST OF ABBREVIATIONS

VGG16	-	Visual Geometry Group 16
CNN	-	Convolutional Neural Networks
Al	-	Artificial Intelligence
ML	-	Machine Learning
ANN	-	Artificial Neural Network
SVM	-	Support Vector Machine
KNN	-	K-Nearest Neighbors
OpenCV	-	Open Source Computer Vision Library
RAM	-	Random Access Memory
RGB	-	Red Green Blue

CHAPTER 1

INTRODUCTION

1.1 DESCRIPTION

Blood group identification is an essential process in the medical field, particularly during blood transfusion, emergency treatment, and surgical procedures. Traditionally, blood group determination requires laboratory testing methods that involve collecting blood samples and performing antigen–antibody reactions. Although these methods provide accurate results, they require medical equipment, trained personnel, and time to perform the analysis. In emergency situations where rapid identification is required, these traditional approaches may cause delays.

Recent advancements in biometric technologies and artificial intelligence have opened new possibilities for predicting biological characteristics using external biometric features. Fingerprints are unique biometric identifiers that contain ridge patterns such as loops, arches, and whorls. These dermatoglyphic patterns have been studied in biomedical research because they may show correlations with certain genetic traits, including blood group types. By analyzing fingerprint patterns using machine learning and deep learning techniques, it may be possible to predict blood group categories without invasive testing.

The project proposes a fingerprint-based blood group prediction system using deep learning models. The system processes fingerprint images using image preprocessing techniques and then applies Convolutional Neural Network (CNN) architectures such as LeNet, AlexNet, VGG16, and ResNet to extract meaningful features. The trained models analyze fingerprint ridge structures and classify them into different blood group categories.

1.2 FUNCTIONAL SPECIFICATION

1.2.1 BIOMETRICS AND FINGERPRINT RECOGNITION

Biometrics refers to the measurement and analysis of unique biological characteristics used for identification and authentication. Fingerprints are one of the most widely used biometric traits due to their uniqueness, permanence, and ease of acquisition. Fingerprint recognition systems analyze ridge patterns, minutiae points, and dermatoglyphic structures to identify individuals. In this project, fingerprint images are used not for identification but for analyzing ridge patterns that may correlate with blood group types.

1.2.2 ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Artificial Intelligence (AI) and Machine Learning (ML) play a significant role in analyzing complex data patterns. Machine learning algorithms can learn relationships between input data and output labels by training on large datasets. In biometric-based healthcare systems, machine learning models can analyze fingerprint images and identify patterns that are difficult for humans to detect. These algorithms help automate the prediction process and improve accuracy through data-driven learning.

1.2.3 DEEP LEARNING FOR IMAGE ANALYSIS

Deep learning is a specialized branch of machine learning that uses neural networks with multiple layers to analyze complex data such as images. Convolutional Neural Networks (CNNs) are particularly effective for image classification tasks because they automatically learn hierarchical features from images.

1.3 IMAGE PROCESSING AND DATA PREPARATION

Image processing plays a crucial role in preparing fingerprint images for deep learning analysis. Raw fingerprint images may contain noise, inconsistent lighting, or varying image sizes, which can negatively affect model performance. Therefore, preprocessing techniques such as resizing, normalization, noise removal, and data augmentation are applied to improve image quality. These preprocessing steps help ensure that the input data is consistent and suitable for training deep learning models. Proper data preparation improves model accuracy and training efficiency.

1.4 OBJECTIVE OF THE PROJECT

The main objective of this project is to develop a deep learning-based system capable of predicting blood groups using fingerprint images. The project aims to analyze fingerprint ridge patterns using CNN architectures and identify correlations between fingerprint features and blood group categories. Another objective is to implement and compare multiple deep learning models such as LeNet, AlexNet, VGG16, and ResNet to evaluate their performance in fingerprint classification tasks.

The system also focuses on improving prediction accuracy through proper image preprocessing, feature extraction, and model training techniques.Additionally, the project aims to demonstrate how biometric identification and artificial intelligence can be integrated to develop automated healthcare systems that support research in non-invasive blood group prediction.

1.5 SCOPE OF THE PROJECT

The scope of this project includes the development of a deep learning framework for fingerprint-based blood group prediction using image classification techniques. The system uses a labeled fingerprint dataset to train CNN models and evaluate their performance in predicting blood groups.

The project mainly focuses on research and experimental implementation of biometric-based blood group prediction methods. It demonstrates the effectiveness of deep learning models in analyzing fingerprint patterns and classifying them into eight blood group categories.

Although the system is designed for academic research, the concept can be extended in the future to develop real-time healthcare applications, biometric identification systems, and automated diagnostic tools. With further research and larger datasets, such systems may contribute to faster and non-invasive blood group identification technologies.

CHAPTER 2

LITERATURE SURVEY

2.1. Automated Blood Group Identification Using Fingerprint Patterns – M. Aravind, P. Keerthana, International Journal of Computer Applications, 2018

The research presented an automated approach for identifying blood groups using fingerprint pattern analysis. The study was based on the concept that fingerprint patterns, which are unique for every individual, could contain biological information related to certain genetic characteristics. The authors aimed to develop a system that could analyze fingerprint images and classify them into different blood group categories. The motivation behind this research was to explore a non-invasive and automated method that could assist in preliminary blood group identification without relying on traditional blood testing procedures. In this work, the researchers focused on analyzing the structural patterns of fingerprints such as loops, arches, and whorls. These ridge patterns were considered important dermatoglyphic features that could be extracted from fingerprint images using image processing techniques. After collecting fingerprint images, preprocessing steps such as image enhancement, noise removal, and segmentation were applied to improve image clarity. Feature extraction techniques were then used to identify the important ridge characteristics that could serve as input data for the classification model. The extracted features helped the system to learn patterns associated with different blood group categories and contributed to the overall effectiveness of the automated prediction system.

2.2. Blood Group Detection Using Fingerprint Image Processing – S. Karthikeyan, R. Rajesh, IEEE Conference on Biomedical Engineering, 2019

The study presented a method for detecting blood groups using fingerprint images combined with image processing and machine learning techniques. The main objective of the research was to explore a non-invasive approach for blood group identification by analyzing fingerprint patterns. Since fingerprints are unique to every individual and remain unchanged throughout life, they were widely used in biometric identification systems. The authors aimed to investigate whether fingerprint characteristics could be used to predict blood group types through automated computational techniques. In the proposed system, the researchers first applied several fingerprint image preprocessing techniques to improve the quality of the input images. Preprocessing was considered an important step because fingerprint images often contained noise, distortions, or unclear ridge patterns. Techniques such as filtering were used to remove noise from the images, while segmentation was applied to isolate the fingerprint region from the background. Ridge enhancement techniques were also used to improve the visibility of fingerprint ridges and valleys so that important fingerprint features could be accurately extracted. After improving the image quality through preprocessing, the researchers used Convolutional Neural Networks (CNN) for feature extraction and classification. CNN was a deep learning technique that was highly effective in image recognition and classification tasks. The results of the study showed that combining fingerprint image processing techniques with deep learning algorithms significantly improved the performance of the prediction system. The CNN model was able to learn complex patterns from fingerprint images and performed classification with better accuracy compared to traditional methods.

2.3. Predicting Blood Group from Fingerprint – D. Rohini, A. Pooja, Sindhu, Journal of Advance Research in Mobile Computing, 2020

The research proposed a system for predicting blood groups by analyzing fingerprint patterns using machine learning techniques. The study aimed to investigate whether fingerprint biometrics could be used as an alternative approach for estimating blood group types. Fingerprints contained unique ridge structures that could be analyzed to extract useful biometric information. The authors focused on developing a system that processed fingerprint images and identified specific features that might be related to different blood group categories. In the proposed method, fingerprint images were first processed to extract important ridge features such as ridge endings, bifurcations, and overall ridge patterns. Feature extraction techniques were used to convert fingerprint images into numerical feature values that could be processed by machine learning algorithms. These extracted features represented the unique characteristics of fingerprints and served as the input data for classification models. Proper preprocessing and feature extraction helped improve the reliability of the classification process. For classification, the researchers implemented two machine learning algorithms: Support Vector Machine (SVM) and Decision Tree. The Decision Tree algorithm was used to identify the most important fingerprint features and understand their relationship with blood group classes. The SVM classifier was then used to perform accurate classification by separating the feature data into different blood group categories. Overall, the system demonstrated that traditional machine learning techniques could effectively analyze fingerprint biometrics for blood group prediction.

2.4. Fingerprint-Based Blood Group Prediction Using Machine Learning – R. V. Jondhale, N. S. Sonawane et al., International Journal of Science, Engineering and Technology, 2021

The study focused on predicting blood groups using fingerprint images through machine learning and deep learning techniques. The research aimed to explore the possibility of identifying blood group types by analyzing fingerprint patterns, which are unique for every individual. Fingerprints contained ridge structures and distinctive patterns such as loops, arches, and whorls that could be analyzed using computational methods. By studying these patterns, the researchers attempted to develop a system that could automatically classify fingerprint images into different blood group categories. In this work, the researchers applied deep learning methods, particularly Convolutional Neural Networks (CNN), to analyze fingerprint images. CNN was widely used in image processing and computer vision tasks because it could automatically learn important features from images without manual feature extraction. Before training the model, several preprocessing techniques were applied to improve the quality of the fingerprint images. These preprocessing steps included image normalization, resizing of images to a fixed dimension, and noise reduction to remove unwanted distortions from the images. After preprocessing, the CNN model was trained using labeled fingerprint datasets associated with known blood groups. The network analyzed the ridge patterns and extracted meaningful features that helped distinguish between different blood group categories. The results of the study demonstrated that deep learning models could effectively classify fingerprint images and support automated blood group prediction systems.

2.5. Blood Group Prediction Using Biometric Fingerprint Analysis – K. S. Praveen, A. Kumar, Springer International Conference on Intelligent Systems, 2022

The research presented a biometric-based approach for predicting blood groups using fingerprint images combined with machine learning techniques. The main objective of the study was to investigate whether fingerprint biometrics could be used as an alternative method for identifying blood group categories. Fingerprints contained unique ridge patterns that remained constant throughout a person's lifetime, making them suitable for biometric analysis. The authors aimed to develop an automated system that could analyze fingerprint images and classify them into different blood group types. In the proposed system, fingerprint images were first processed using various image processing techniques to extract important fingerprint features. These techniques helped identify ridge patterns, bifurcations, and other structural characteristics present in fingerprint images. The extracted features were then converted into numerical data that could be used as input for machine learning algorithms. Proper preprocessing and feature extraction played a significant role in improving the accuracy and reliability of the prediction system. For the classification stage, the researchers implemented the Support Vector Machine (SVM) algorithm. SVM was a powerful machine learning technique widely used for classification problems because it could efficiently handle high-dimensional feature data. The SVM model analyzed the extracted fingerprint features and separated them into different blood group categories using decision boundaries.

2.6. Fingerprint-Based Blood Group Classification Using Deep Learning – Marriya Tabassum & Dr. Khaja, International Journal of Innovative Research in Engineering, 2022

The study proposed a fingerprint-based blood group classification system using machine learning techniques. The main objective of the research was to analyze whether fingerprint biometric patterns could be used to predict blood group categories. Fingerprints contained unique ridge structures such as loops, arches, and whorls, which could be analyzed using computational methods. The researchers developed a system that processed fingerprint images and extracted important biometric features that might be related to different blood group types. In the proposed approach, fingerprint images were first processed using feature extraction techniques to identify important ridge structures and fingerprint patterns. These features were converted into numerical data that could be used as input for machine learning algorithms. After feature extraction, several classification algorithms such as K-Nearest Neighbors (KNN), Naïve Bayes, and Random Forest were implemented to analyze the fingerprint data. Each algorithm was trained using labeled fingerprint datasets associated with known blood groups. Among the applied algorithms, Random Forest helped improve prediction accuracy by combining multiple decision trees and producing more reliable classification results. The Naïve Bayes classifier performed probabilistic classification based on the likelihood of fingerprint features belonging to a particular blood group category. The experimental results showed that using multiple machine learning algorithms enhanced the performance of biometric classification systems and improved the accuracy of blood group prediction using fingerprint images.

2.7. Innovative Blood Group Prediction Using Fingerprint Patterns – Dr. Senthilkumar R., Mohamed Harish S., Ragul M., Navinkumar A., International Journal of Engineering Research & Technology (IJERT), 2023

The research focused on predicting blood groups using fingerprint pattern recognition techniques. The study was based on the concept that dermatoglyphic features of fingerprints, such as loops, whorls, and arches, might show certain correlations with genetic characteristics including blood group types. Fingerprints were unique for every individual and remained unchanged throughout life, making them an important biometric trait for identification and analysis. The authors aimed to investigate whether these fingerprint patterns could be used to develop a system capable of predicting blood groups in a non-invasive manner. In the proposed system, fingerprint images were first processed using several image preprocessing techniques to improve image quality and ensure accurate feature extraction. Techniques such as segmentation were used to isolate the fingerprint region from the background, while filtering methods helped remove noise and improved the clarity of ridge patterns. Pattern extraction methods were then applied to identify important dermatoglyphic features such as ridge flow, loops, whorls, and arches present in the fingerprint images. After extracting the relevant fingerprint features, machine learning algorithms were used to classify the fingerprints into different blood group categories. The system analyzed the extracted features and predicted the corresponding blood group types. The results of the study demonstrated that fingerprint pattern recognition techniques could be effectively used for non-invasive blood group prediction.

2.8. Blood Group Detection Using Fingerprint Images – Vidya Waykule, Sakshi Amrutkar, Om Jadhav, Vardhman Jain, Titiksha Jawale, International Journal for Research in Applied Science and Engineering Technology, 2024

The study presented a non-invasive approach for detecting blood groups using fingerprint images combined with machine learning techniques. The main aim of the research was to develop an automated system that could analyze fingerprint patterns and predict blood group types without the need for traditional blood sample testing. Fingerprints were widely used in biometric identification because of their uniqueness and permanence. By studying the ridge patterns and structures present in fingerprints, the researchers attempted to identify possible relationships between fingerprint features and blood group categories. In the proposed system, fingerprint images were first processed using image enhancement techniques to improve their quality and clarity. Image enhancement methods such as filtering, normalization, and noise reduction were applied to remove distortions and highlight important ridge patterns. After enhancing the images, feature extraction techniques were used to identify key biometric characteristics from the fingerprints, including ridge patterns, texture information, and structural details. These extracted features were then converted into numerical values that could be used as input for machine learning models. The extracted fingerprint features were analyzed using machine learning classification algorithms to predict the corresponding blood group. The results of the study demonstrated that combining image processing with machine learning techniques could effectively support non-invasive blood group prediction.

2.9. Deep Learning-Based Blood Group Prediction Using Fingerprint Biometrics – Abhishek Tripathi, Palukuru, Springer (ICRT CIS), 2025

The research explored the use of deep learning and machine learning algorithms for predicting blood groups using fingerprint biometric data. The main objective of the study was to develop an intelligent system capable of analyzing fingerprint patterns and identifying the corresponding blood group categories. Fingerprints contained distinctive ridge patterns and structural characteristics that could be analyzed using computational techniques. By using machine learning and deep learning approaches, the researchers aimed to automatically learn meaningful patterns from fingerprint images and improve the efficiency of the prediction process. In this study, several machine learning algorithms were implemented, including Artificial Neural Networks (ANN), K-Nearest Neighbors (KNN), and Gradient Boosting. Artificial Neural Networks were used to recognize complex fingerprint patterns and learn relationships between fingerprint features and blood group categories. The K-Nearest Neighbors algorithm classified fingerprint images based on similarity measures between extracted features. By comparing new fingerprint data with previously labeled samples, the KNN algorithm helped determine the most likely blood group classification. To further enhance the prediction accuracy, the researchers applied Gradient Boosting techniques, which combined multiple weak learning models to produce stronger classification results. Feature normalization techniques were also used to standardize the extracted data and improve the stability and performance of the machine learning models. The results of the study demonstrated that combining different machine learning algorithms with proper feature preprocessing significantly improved the accuracy of fingerprint-based blood group prediction systems.

2.10. A Machine Learning-Based Non-Invasive System for Blood Group Prediction Using Fingerprint Biometrics – Ritika Yadav, Saubhagya, Research & Reviews Journal of Bioinformatics, 2025

The research introduced a machine learning-based system for predicting blood groups using fingerprint biometric data. The primary objective of the study was to develop an automated system that could analyze fingerprint images and identify the corresponding blood group category. Fingerprints contained unique ridge patterns that could be processed using image analysis techniques to extract meaningful biometric features. By combining biometric identification with machine learning models, the researchers aimed to create a system that could assist in predicting blood groups in a non-invasive and efficient manner. In the proposed system, several technologies were integrated to process and analyze fingerprint images. Image processing techniques were implemented using OpenCV to enhance fingerprint quality, remove noise, and extract important ridge features. The machine learning models were developed using Python and TensorFlow to train the system with labeled fingerprint datasets. These models learned the relationship between fingerprint patterns and blood group categories through repeated training and feature analysis. The trained model was then capable of classifying new fingerprint images into their respective blood group types. To make the system more accessible and user-friendly, the researchers developed a web-based interface using React for the frontend and Flask for the backend. This interface allowed users to upload fingerprint images and obtain blood group predictions from the trained machine learning model.

CHAPTER 3

EXISTING SYSTEM

3.1 TRADITIONAL BLOOD GROUP DETECTION METHODS

The existing system for blood group detection mainly relies on traditional laboratory testing methods where blood samples are collected from individuals and analyzed using chemical reagents. This process identifies blood groups based on the presence or absence of specific antigens and antibodies in the blood. Although this method is highly accurate and widely used in hospitals and medical laboratories, it is invasive and requires specialized equipment, trained personnel, and sufficient time for analysis. In emergency situations or rural areas where medical facilities are limited, performing such tests quickly can be difficult.

To overcome these limitations, researchers have explored the use of biometric features such as fingerprints for blood group prediction. Fingerprint images are used as input data and analyzed using machine learning and deep learning techniques. The system uses a dataset containing thousands of fingerprint images labeled with known blood group types such as A+, A−, B+, B−, AB+, AB−, O+, and O−. These images are processed and used to train deep learning models that can recognize patterns in fingerprint ridges and associate them with blood group categories.

The existing system applies Convolutional Neural Network (CNN) architectures such as ResNet, VGG16, AlexNet, and LeNet to extract features from fingerprint images and perform classification. These deep learning models automatically learn complex fingerprint patterns through training and then classify new fingerprint images into corresponding blood groups. The project structure includes datasets, training code, performance graphs, and testing modules to evaluate the models and compare their accuracy.

Although the existing system demonstrates that fingerprint-based prediction is possible using deep learning models, it still has some limitations. The accuracy of prediction depends heavily on the size and quality of the dataset, and the scientific correlation between fingerprint patterns and blood groups is still under research. Therefore, the existing system is mainly considered a research-based or experimental approach rather than a replacement for standard medical blood group testing methods. The existing diagram is implemented in figure 3.1.

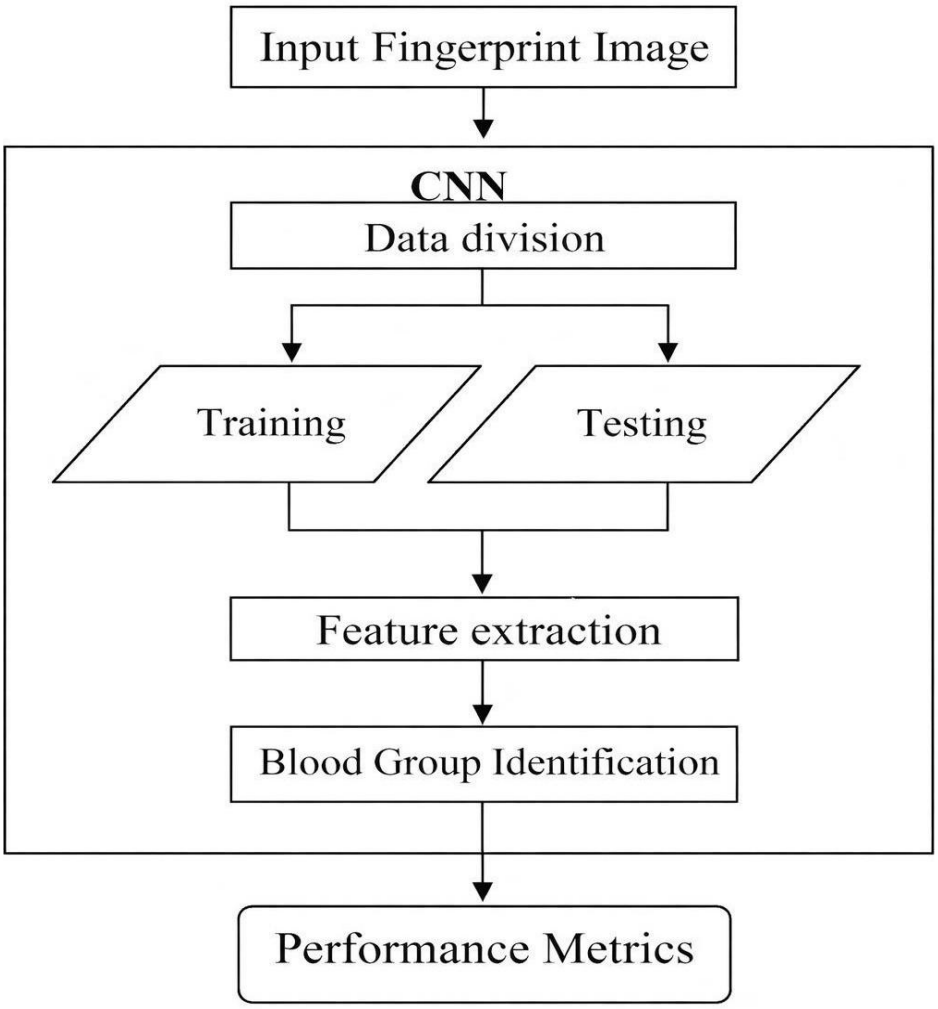


Figure 3.1. Existing Block Diagram

CHAPTER 4

PROBLEMS IDENTIFIED

4.1 IDENTIFICATION OF SYSTEM CHALLENGES

The development of a fingerprint-based blood group detection system involves several challenges related to data collection, image quality, model accuracy, and practical implementation. While deep learning provides powerful tools for image classification, the effectiveness of the system depends on multiple technical and data-related factors. During the analysis and development of this project, several key problems were identified that influence the performance and reliability of the system.

One of the primary problems identified is the availability and quality of fingerprint datasets associated with blood group labels. In many biometric datasets, fingerprint images are available, but they are not always labeled with the corresponding blood group information. This creates difficulties in obtaining large, reliable datasets required for training deep learning models. Limited dataset size may reduce the ability of the model to learn complex fingerprint patterns associated with different blood groups.

Another important issue is variation in fingerprint image quality. Fingerprint images collected from different sources may contain noise, distortions, uneven lighting, or incomplete ridge structures. Poor image quality can negatively affect feature extraction and reduce the accuracy of the classification models. Therefore, effective preprocessing techniques are required to standardize the images and enhance the ridge patterns before feeding them into the deep learning model. The complex relationship between fingerprint patterns and blood groups is also a challenge in this research area.

Although dermatoglyphic studies suggest possible correlations between fingerprint patterns and blood groups, the relationship is not always strong or consistent. As a result, machine learning models may struggle to identify clear patterns that reliably predict blood group categories. Another problem identified is the selection of appropriate deep learning architectures and hyperparameters. Different CNN models such as LeNet, AlexNet, VGG16, and ResNet have different complexities and computational requirements. Choosing the right architecture, learning rate, batch size, and training parameters is necessary to achieve optimal model performance. Improper configuration may lead to problems such as overfitting, underfitting, or slow training.

Computational resource limitations also present challenges during model training. Deep learning models require significant processing power and memory, especially when training large datasets. Systems with limited hardware resources may experience longer training times and reduced efficiency, which can affect experimentation and model optimization.

Finally, there are practical implementation and reliability concerns when applying fingerprint-based blood group prediction systems in real-world healthcare environments. Since blood group identification is a critical medical process, any prediction system must achieve very high accuracy and reliability. Further research, larger datasets, and extensive validation are required before such systems can be used in real clinical applications.

Overall, the problems identified in this project highlight the need for improved datasets, better image preprocessing methods, optimized deep learning models, and further research to enhance the reliability of fingerprint-based blood group prediction systems.

CHAPTER 5

PROPOSED SYSTEM

5.1 OVERVIEW OF PROPOSED SYSTEM

The proposed system introduces an automated and non-invasive method for predicting blood groups using fingerprint images and deep learning techniques. Instead of collecting blood samples and performing laboratory tests, the system uses fingerprint biometrics as the input data. Fingerprint images are analyzed using machine learning algorithms to identify patterns that may correspond to specific blood group categories. The goal of the proposed system is to create a faster, cost-effective, and easily accessible method for preliminary blood group prediction using artificial intelligence.

In the proposed approach, a dataset containing thousands of fingerprint images labeled with their respective blood groups is used for training the model. Each fingerprint image undergoes preprocessing steps such as resizing, normalization, and noise removal to improve image quality and ensure accurate feature extraction. After preprocessing, deep learning models analyze the fingerprint ridge patterns and extract meaningful features automatically. The system then uses these features to classify the fingerprint into one of the eight blood group categories: A+, A−, B+, B−, AB+, AB−, O+, and O−.

The proposed system implements several Convolutional Neural Network (CNN) architectures such as ResNet, VGG16, AlexNet, and LeNet. These deep learning models are trained using fingerprint datasets to learn complex patterns present in fingerprint images. After training, the models are evaluated using testing datasets, and performance metrics such as accuracy, validation accuracy, and loss are generated. The project also compares the performance of different models to determine which architecture provides better prediction accuracy for blood group classification.

The final system is designed to take a fingerprint image as input and automatically predict the corresponding blood group as output. The trained model can be tested using new fingerprint images stored in the testing folder. The results are visualized using performance graphs and comparison charts to analyze model efficiency. This proposed system demonstrates how biometric identification combined with deep learning can be used to develop intelligent healthcare prediction systems, although it is primarily intended for research and academic purposes. The proposed diagram is implemented in figure 5.1.

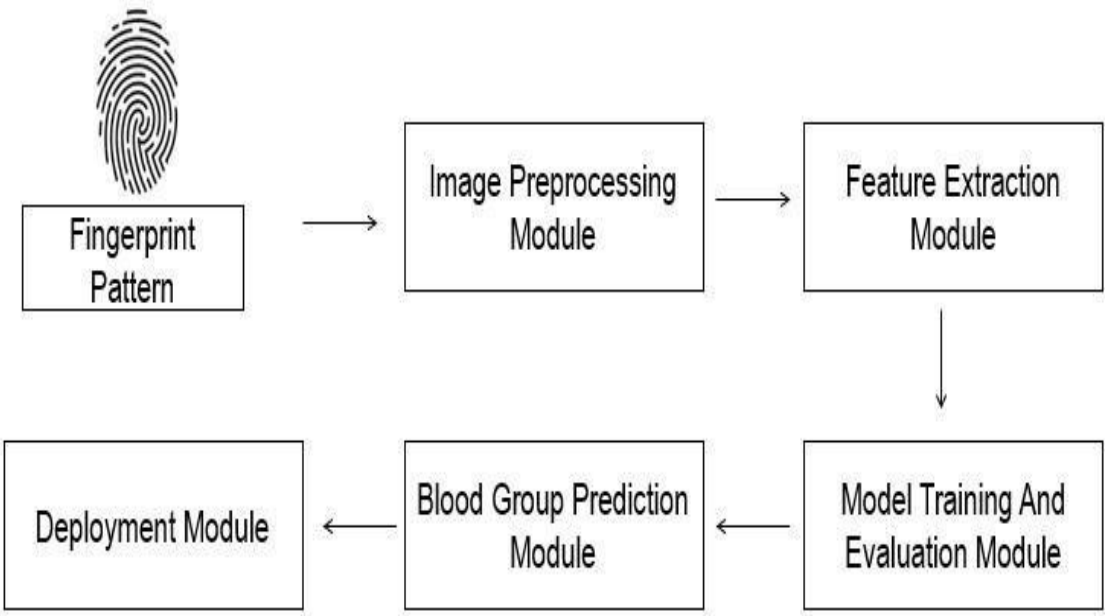


Figure 5.1 Proposed Architecture Diagram

CHAPTER 6

SYSTEM REQUIREMENTS

6.1 HARDWARE REQUIREMENTS

COMPONENT	SPECIFICATION
Processor	Intel Core i3 or higher
RAM	Minimum 4 GB (8GB Recommended)
Storage	10 GB free disk space
Fingerprint Input	Fingerprint image samples or dataset
Display	Standard Monitor (1024x768 resolution)

The hardware requirements for the system include a computer with at least an Intel Core i3 processor or higher to ensure smooth execution of the deep learning models and image processing tasks. A minimum of 4 GB RAM is required, although 8 GB is recommended for better performance during training and testing processes. The system should have at least 10 GB of free storage space to accommodate the fingerprint dataset, trained models, and related files. Fingerprint input is provided in the form of image samples or datasets, which serve as the primary data source for the system. Additionally, a standard monitor with a resolution of 1024×768 or higher is sufficient for displaying outputs and interacting with the system interface.

6.2 SOFTWARE REQUIREMENTS

COMPONENT	SPECIFICATION
Operating System	Windows 10 / Linux /macOS
Programming Language	Python
Python Libraries	Opencv, numpy, Pillow, Torchvision, PyTorch
Webcam Driver	Standard camera drivers with OpenCV
Deep Learning Models	Torchvision(AlexNet, VGG16, ResNet, LeNet)
Dataset	Fingerprint image dataset with blood group labels
IDE / Editor	Visual Studion Code

The software requirements for the system include an operating system such as Windows 10, Linux, or macOS to support development and execution. The primary programming language used is Python, along with essential libraries such as OpenCV, NumPy, Pillow, Torchvision, and PyTorch for image processing and deep learning tasks. Standard webcam drivers integrated with OpenCV can be used if real-time image input is required.

CHAPTER 7

SYSTEM IMPLEMENTATION

7.1 SYSTEM IMPLEMENTATION OVERVIEW

The system implementation focuses on developing an end-to-end deep learning framework for predicting blood groups from fingerprint images by integrating multiple processing stages into a unified workflow. Initially, fingerprint images are collected and prepared as input data, followed by the application of image preprocessing techniques to enhance quality, remove noise, and ensure consistency across the dataset. These features are utilized to train various convolutional neural network architectures, enabling the system to learn complex relationships between fingerprint characteristics and blood group categories. After training and evaluation, the optimized model is used to predict blood groups from new fingerprint images. The final system is designed to provide an efficient, automated, and non-invasive solution, and can be further deployed through user-friendly applications for real-time prediction.

7.2 LIST OF MODULES

The system is divided into several modules, each responsible for a specific function in the overall workflow. These modules work together sequentially to process fingerprint images, extract meaningful features, train the deep learning models, and generate accurate blood group predictions.

- Fingerprint Image Acquisition Module
- Image Preprocessing Module
- Feature Extraction Module
- Model Training and Evaluation Module
- Blood Group Prediction and Deployment Module

7.3 MODULES DESCRIPTION

The proposed fingerprint-based blood group detection system is divided into five main modules that work together to perform accurate prediction. The Fingerprint Image Acquisition Module is responsible for collecting fingerprint images from a dataset or biometric source, where each image is labeled according to its corresponding blood group category. After acquiring the images, the Image Processing Module prepares the raw fingerprint images for analysis by applying preprocessing techniques such as resizing, normalization, noise removal, and image enhancement to improve image quality and maintain consistency.

The processed images are then passed to the Feature Extraction Module, where deep learning models such as Convolutional Neural Networks (CNNs) automatically identify important fingerprint characteristics like ridge patterns, edges, loops, arches, and whorls that are useful for classification. These extracted features are used in the Model Training and Evaluation Module, where different CNN architectures such as LeNet, AlexNet, VGG16, and ResNet are trained using the labeled dataset. During this stage, the models learn patterns related to different blood groups and their performance is evaluated using metrics such as accuracy, loss, and validation results.

Finally, the Blood Group Prediction and Deployment Module uses the trained model to predict the blood group from new fingerprint images by analyzing the learned patterns and classifying them into one of the eight blood group categories. This module also enables the system to be deployed for practical use through a user interface or application for real-time prediction.

7.3.1 FINGERPRINT IMAGE ACQUISITION MODULE

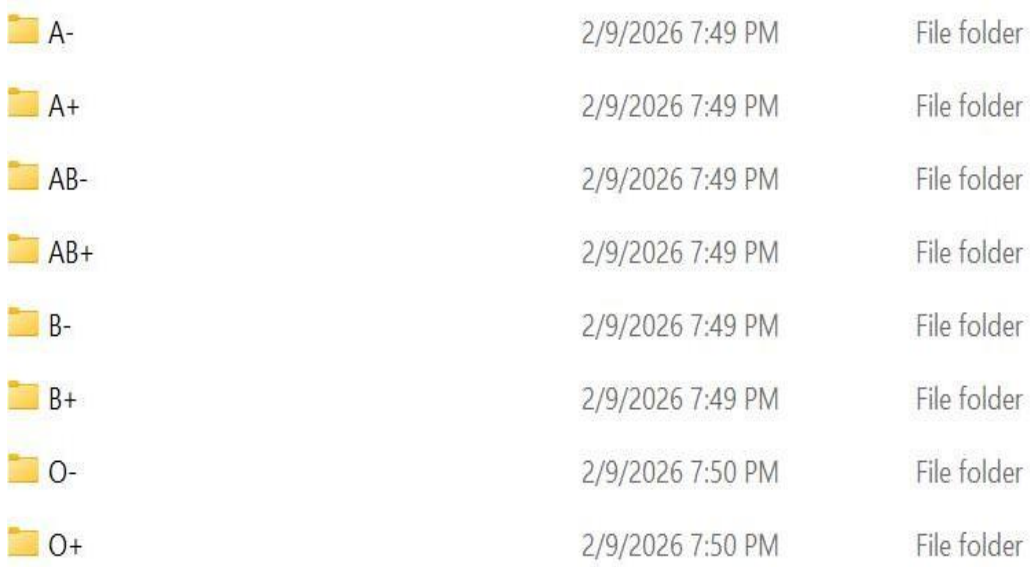
The Fingerprint Image Acquisition Module is responsible for collecting and preparing fingerprint images that will be used as input for the blood group prediction system. Fingerprints are one of the most widely used biometric identifiers because every individual has unique ridge patterns. In this system, fingerprint images serve as the primary data source that allows the machine learning model to analyze patterns and make predictions. The collected images form the base dataset on which the entire prediction system is built.

In this project, fingerprint images are obtained from a labeled dataset in which each fingerprint sample is associated with a specific blood group. The blood group categories include A+, A−, B+, B−, AB+, AB−, O+, and O−. Each image in the dataset is assigned to one of these categories, allowing the machine learning model to learn the relationship between fingerprint patterns and blood group types during the training phase. The labeled dataset helps the model understand which fingerprint features correspond to particular blood groups.

Fingerprint images can be collected through various methods such as biometric fingerprint scanners, mobile fingerprint sensors, or publicly available fingerprint datasets used for research purposes. When a fingerprint image is captured, it contains detailed ridge structures that form unique patterns. These patterns include loops, arches, and whorls, which are commonly studied in the field of dermatoglyphics.

The collected fingerprint images are usually organized in a structured dataset format. In this structure, images belonging to the same blood group category are stored in separate folders or labeled directories. It also allows the deep learning models to automatically read and process images from different categories.

Overall, the fingerprint image acquisition stage plays a crucial role in the success of the entire system. The accuracy and reliability of the blood group prediction model largely depend on the quality, clarity, and diversity of the collected fingerprint images. The fingerprint image acquisition module is implemented in figure 7.2.1.



A-	2/9/2026 7:49 PM	File folder
A+	2/9/2026 7:49 PM	File folder
AB-	2/9/2026 7:49 PM	File folder
AB+	2/9/2026 7:49 PM	File folder
B-	2/9/2026 7:49 PM	File folder
B+	2/9/2026 7:49 PM	File folder
O-	2/9/2026 7:50 PM	File folder
O+	2/9/2026 7:50 PM	File folder

Figure 7.3.1. Fingerprint Image Acquisition Module

7.3.2 IMAGE PREPROCESSING MODULE

The Image Preprocessing Module plays an important role in preparing raw fingerprint images for further analysis by the machine learning model. Fingerprint images collected from sensors or datasets often contain various imperfections such as noise, low contrast, uneven lighting, or blurred ridge structures. These issues can affect the ability of the deep learning model to correctly identify fingerprint features.

Therefore, preprocessing techniques are applied to improve the quality of the images before they are used for feature extraction and model training. In this stage, several image processing techniques are used to standardize the fingerprint images. One of the first steps is image resizing, where all fingerprint images are converted into a fixed resolution. This ensures that every image fed into the neural network has the same dimensions, which is necessary for consistent training. After resizing, the images are converted into grayscale format, which simplifies the image data by reducing it from three color channels (RGB) to a single intensity channel. This step reduces computational complexity and focuses the analysis on the structural patterns of the fingerprint.

Another important preprocessing step is noise removal. Fingerprint images may contain random noise caused by sensor errors, environmental conditions, or image compression. Noise removal filters such as smoothing or median filtering are applied to remove unwanted disturbances from the images. This helps to produce clearer ridge patterns and improves the reliability of feature detection during later stages of the system. In addition to noise removal, image normalization and contrast enhancement techniques are used to improve the clarity of fingerprint ridges and valleys.

Normalization adjusts the pixel intensity values so that the brightness and contrast of all images remain consistent. Contrast enhancement helps highlight the fingerprint ridge structures, making them more visible for feature extraction. These improvements make it easier for the deep learning model to learn meaningful fingerprint patterns. The image preprocessing module is implemented in figure 7.3.2.

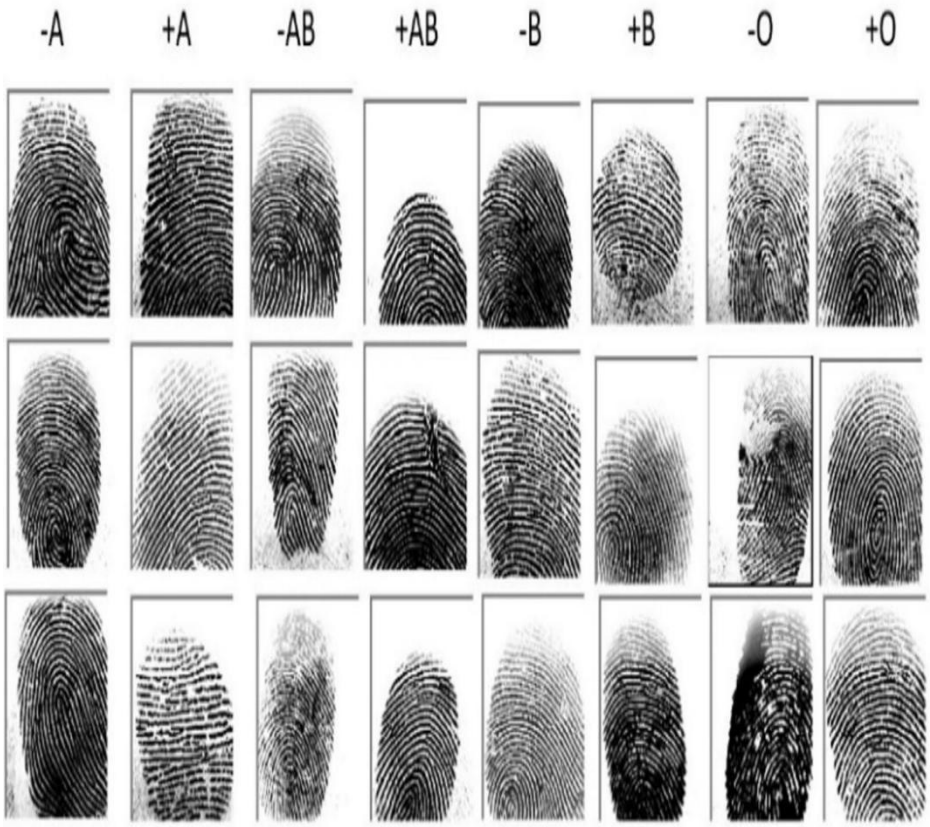


Figure 7.3.2 Image Preprocessing Module

7.3.3 FEATURE EXTRACTION MODULE

The Feature Extraction Module is responsible for identifying important characteristics from the preprocessed fingerprint images. After the images have been cleaned and enhanced in the preprocessing stage, the system analyzes them to detect meaningful fingerprint patterns. Fingerprints contain several unique structures such as ridge endings, bifurcation points, loops, arches, and whorls. These patterns form the basis of biometric identification and can provide valuable information that helps in classification tasks.

In fingerprint analysis, these ridge patterns represent the unique biological structure of a person's fingerprint. Ridge endings occur where a ridge line stops, while bifurcation points occur where a ridge splits into two branches. Other patterns such as loops, arches, and whorls represent the general shape of fingerprint ridges. These dermatoglyphic features are widely used in biometric systems because they remain stable throughout a person's life and vary from one individual to another.

In traditional fingerprint recognition systems, feature extraction is usually performed using manual algorithms that detect specific points known as minutiae points. These algorithms analyze the ridge structures in the fingerprint image and locate important points such as ridge endings and bifurcations. The detected features are then converted into numerical values that can be used for classification or identification purposes.

However, in this project, feature extraction is performed automatically using deep learning techniques, particularly Convolutional Neural Networks (CNN). CNN models are designed to analyze image data by passing it through multiple convolutional layers. The initial layers of the CNN detect simple features such as edges, lines, and textures, while deeper layers identify more complex patterns such as ridge shapes and fingerprint structures.

This hierarchical learning process allows the system to capture detailed fingerprint features without manual intervention.The automatic feature extraction capability of CNN significantly improves the efficiency and accuracy of the system. Since the model learns important patterns directly from the training data, it eliminates the need for manual feature engineering. The extracted features are then forwarded to the classification stage, where the machine learning model analyzes them and predicts the most probable blood group category.

7.3.4 MODEL TRAINING AND EVALUATION MODULE

The Model Training and Evaluation Module is responsible for training the deep learning models using the features extracted from fingerprint images. After the feature extraction stage, the processed fingerprint data is used to train machine learning models so that they can learn the relationship between fingerprint patterns and blood group categories. This module plays a key role in enabling the system to recognize patterns and make accurate predictions based on the training data.

In this system, several Convolutional Neural Network (CNN) architectures are implemented to perform the classification task. The models used include ResNet, VGG16, AlexNet, and LeNet, which are well-known deep learning architectures for image recognition problems. Each of these models has different network depths and structures that allow them to learn complex image features. By training multiple models, the system can compare their performance and determine which architecture provides the best prediction accuracy for the fingerprint dataset.

During the training process, the models analyze thousands of fingerprint images from the labeled dataset. The dataset is typically divided into training and validation sets. The training set is used to teach the model how to recognize fingerprint patterns associated with different blood groups, while the validation set is used to check how well the model performs on unseen data. The model adjusts its internal parameters such as weights and biases through multiple iterations using optimization algorithms to minimize classification errors. Several performance metrics are used to monitor the learning progress of the models during training. These metrics include training accuracy, validation accuracy, training loss, and validation loss.

7.3.5 BLOOD GROUP PREDICTION AND DEPLOYMENT MODULE

The Blood Group Prediction and Deployment Module is the final stage of the fingerprint-based blood group detection system. In this module, the trained deep learning model is used to predict the blood group of a person using a new fingerprint image as input. Once the model has completed its training and evaluation phases, it is ready to process unseen fingerprint samples. The system analyzes the fingerprint image and provides the predicted blood group category as the output.

In this stage, a fingerprint image is provided to the system as input. The input image first goes through the same preprocessing steps used during the training stage. These steps include resizing the image, converting it into grayscale, and improving image quality through noise removal and normalization. This ensures that the input image has the same format and quality as the images used to train the model.

After preprocessing, the image moves to the feature extraction stage, where the trained convolutional neural network analyzes the fingerprint ridge patterns. The CNN automatically detects important fingerprint features such as ridge structures, loops, arches, and whorls. These features are converted into numerical feature representations that the model can use for classification. Once the features are extracted, the trained classification model processes the information and predicts the most probable blood group category. The system selects one of the eight possible classes, which include A+, A−, B+, B−, AB+, AB−, O+, and O−. The predicted blood group is then displayed as the final output of the system. This automated process requires only a fingerprint image as input, making the system simple and efficient to use.

CHAPTER 8

SYSTEM TESTING

System testing ensures that the fingerprint-based blood group detection system works correctly and efficiently. The testing process verifies whether all modules such as fingerprint image acquisition, preprocessing, feature extraction, model training, and prediction operate properly together. Various testing methods are used to check the functionality, performance, security, and usability of the system. These tests help identify errors, validate the prediction results, and improve the reliability of the machine learning model.

8.1. UNIT TESTING

Unit testing is a software testing technique that focuses on verifying the functionality of individual components or modules of a system independently. The main goal of unit testing is to ensure that each module performs its intended task correctly without depending on other parts of the system. By testing small units of the program separately, developers can easily identify errors or bugs at an early stage of development. This approach improves the reliability and quality of the system before the modules are combined into a complete application.

In this project, unit testing is applied to several important modules such as the image preprocessing module, feature extraction module, and model prediction module. Each of these modules performs a specific function in the fingerprint-based blood group detection system. During unit testing, the functions written in Python are tested independently to verify that they operate correctly when processing fingerprint images. This ensures that every stage of the system performs its assigned task properly.

For example, the image preprocessing module is tested to confirm that fingerprint images are correctly resized, converted to grayscale, normalized, and cleaned from noise. Similarly, the feature extraction module is tested to verify that the convolutional neural network correctly identifies fingerprint patterns and extracts meaningful features from the images. The model prediction module is also tested to ensure that the trained model receives the correct input features and generates the expected blood group prediction output. The unit testing is implemented in figure 8.1.

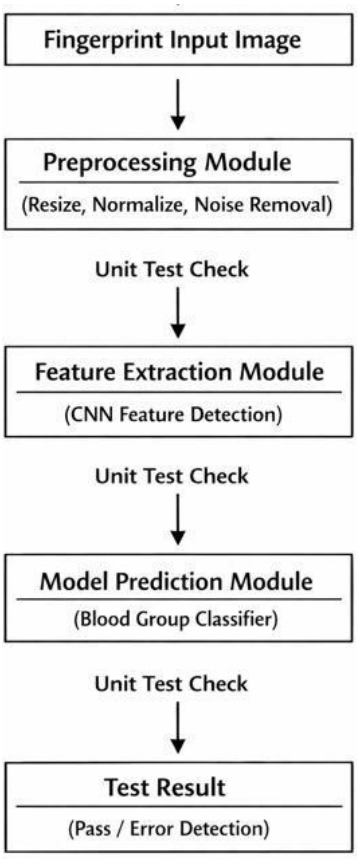


Figure 8.1 Unit Testing

8.2. INTEGRATION TESTING

Integration testing is a software testing technique used to verify whether different modules of a system work correctly when combined together. After individual modules are tested through unit testing, integration testing ensures that these modules interact properly and function as a complete system. The main objective of integration testing is to detect errors that may occur when different components communicate or exchange data with each other.

In the fingerprint-based blood group detection project, several modules such as fingerprint image acquisition, image preprocessing, feature extraction, and deep learning classification are integrated to form a complete processing pipeline. During integration testing, these modules are connected sequentially so that the output produced by one module becomes the input for the next module. This integration ensures that the workflow of the system operates smoothly from the initial fingerprint input stage to the final prediction stage.

During this testing phase, fingerprint images are passed through the entire system pipeline starting from preprocessing, followed by feature extraction, and finally classification by the trained model. Integration testing verifies that data flows correctly between modules and that no errors occur during data transfer or processing. This process confirms that the integrated system functions properly and consistently produces correct prediction results when all modules operate together.

8.3. SYSTEM TESTING

System testing is a software testing process used to evaluate the entire system as a complete and integrated unit. The main objective of system testing is to verify that the developed system meets all the specified functional and performance requirements.

Unlike unit testing and integration testing, which focus on individual modules or their interactions, system testing checks the behavior of the full application in a real working environment. This ensures that all components of the system operate together correctly and produce the expected results. In the fingerprint-based blood group detection project, system testing is performed after all modules of the system have been successfully integrated. During this phase, fingerprint images from the testing dataset are provided as input to the trained deep learning models.

These images represent real test cases used to evaluate how well the system performs when predicting blood groups based on fingerprint patterns. When the system receives a fingerprint image, it processes the image through several stages including image preprocessing, feature extraction, and classification. The preprocessing stage improves the quality of the fingerprint image, while the feature extraction stage identifies important ridge patterns and biometric features. Finally, the trained deep learning model analyzes these extracted features and predicts the corresponding blood group category.

The predicted results generated by the system are then compared with the actual blood group labels in the dataset to evaluate the performance of the model. Metrics such as prediction accuracy and classification correctness are used to measure the effectiveness of the system. System testing confirms that the overall system works correctly and demonstrates that the model can successfully predict blood groups from fingerprint images in a complete end-to-end workflow.

In addition to accuracy evaluation, system testing also ensures that the system performs reliably under different conditions and inputs. Various test cases are executed using fingerprint images with different qualities, patterns, and variations to verify the robustness of the system.

8.4. PERFORMANCE TESTING

Performance testing is used to evaluate how efficiently and reliably a system performs under different operating conditions. The main objective of performance testing is to measure the speed, responsiveness, and stability of the system when it processes data. In machine learning and deep learning projects, performance testing is particularly important because it helps determine how well the models perform in terms of accuracy, training time, and prediction speed.

In the fingerprint-based blood group detection project, performance testing mainly focuses on evaluating the training and prediction efficiency of the deep learning models used in the system. During this stage, different models are trained using the fingerprint dataset, and their performance is analyzed based on several evaluation metrics. These metrics help determine how effectively the models learn fingerprint patterns and classify them into the correct blood group categories.

Important performance metrics such as training accuracy, validation accuracy, loss values, and model training time are carefully analyzed. Accuracy represents how correctly the model predicts blood group categories, while loss values indicate the error rate during training. Graphs showing changes in accuracy and loss over multiple training epochs are generated to visualize the learning progress of the models. These graphs help identify whether the model is improving during training or facing issues such as overfitting or underfitting. Different Convolutional Neural Network architectures such as AlexNet, VGG16, ResNet, and LeNet are compared during performance testing. Each model has different architectural complexity and computational requirements.

8.5. SECURITY TESTING

Security testing is performed to ensure that the system protects sensitive data and prevents unauthorized access. The main objective of security testing is to identify potential vulnerabilities in the system that could lead to data breaches or misuse of information. In biometric-based systems, security becomes very important because the data being handled, such as fingerprint images, is unique to individuals and cannot be easily replaced if compromised. In the fingerprint-based blood group detection project, fingerprint images are used as biometric input data for training and testing the deep learning models. Since fingerprint data is highly sensitive, it is necessary to ensure that the dataset is handled securely throughout the system. Security testing verifies that the fingerprint dataset is stored safely and is accessed only by authorized users for research and model development purposes.

Another important aspect of security testing in this project is controlling access to the trained models and prediction system. The system ensures that only authorized users or developers can modify the model, access the dataset, or run the training process. Proper access control helps prevent unauthorized modifications or misuse of the biometric data and trained models.

In addition, the system ensures that fingerprint images provided as input are processed only within the secure prediction environment. The images are used only for analysis and prediction, and sensitive biometric information is not exposed or shared outside the system. Through these security measures, the system maintains privacy and ensures that biometric data is handled according to proper data protection standards.

CHAPTER 9

RESULT AND DISCUSSION

9.1 RESULTS AND DISCUSSION ANALYSIS

The proposed Fingerprint-Based Blood Group Detection System was implemented using deep learning techniques to analyze fingerprint images and predict the corresponding blood group. The system was trained using a labeled fingerprint dataset containing images categorized according to different blood group classes. Several preprocessing steps such as image resizing, normalization, and noise reduction were applied to improve the quality of the fingerprint images before training the models. After preprocessing, the images were used as input for different Convolutional Neural Network (CNN) architectures including AlexNet, VGG16, ResNet, and LeNet. These models were trained to learn the unique fingerprint ridge patterns and classify them into blood group categories.

During the training phase, the models were evaluated using metrics such as training accuracy, validation accuracy, and loss values. Graphs representing the training and validation performance were generated to analyze the learning behavior of the models. The results showed that the deep learning models were able to extract meaningful fingerprint features and gradually improve their classification performance during training. Among the tested architectures, deeper models such as ResNet and VGG16 generally demonstrated better performance due to their ability to capture complex image features.

The experimental results indicate that the proposed system can successfully predict blood groups based on fingerprint images with reasonable accuracy. The comparison between different CNN models helped identify the most efficient architecture for the task.

The performance evaluation also showed that proper preprocessing and feature extraction significantly improved model accuracy. This demonstrates the importance of high-quality data preparation in machine learning-based biometric systems. Overall, the results confirm that fingerprint biometrics combined with deep learning algorithms can be used to develop a non-invasive approach for blood group prediction. Although the system provides promising results, further improvements such as using larger datasets, advanced deep learning models, and real-time deployment can enhance the reliability and accuracy of the system. The study highlights the potential of artificial intelligence in developing innovative healthcare applications that reduce dependency on traditional laboratory testing methods.

CHAPTER 10

10.1 CONCLUSION

The Fingerprint-Based Blood Group Detection System demonstrates the potential of using biometric data and deep learning techniques to predict blood groups in a non-invasive manner. Traditional blood group identification methods require laboratory testing and blood samples, which may take time and require medical equipment. In contrast, the proposed system utilizes fingerprint images as input and applies deep learning algorithms to analyze fingerprint ridge patterns and predict the corresponding blood group category. This approach highlights how artificial intelligence can be integrated with biometric technologies to develop innovative healthcare solutions.

In this project, fingerprint images are processed through several stages including image acquisition, preprocessing, feature extraction, model training, and prediction. Deep learning architectures such as AlexNet, VGG16, ResNet, and LeNet are used to analyze fingerprint patterns and classify them into different blood group types. The system evaluates the performance of these models using metrics such as accuracy and loss, allowing comparison between different neural network architectures. The results indicate that deep learning models can effectively learn complex fingerprint features and perform classification tasks with reasonable accuracy.

The project also demonstrates the importance of proper dataset preparation, preprocessing techniques, and model evaluation in machine learning applications. By combining image processing techniques with convolutional neural networks, the system is able to extract meaningful patterns from fingerprint images and use them for prediction. The implementation using Python, PyTorch, and Google Colab provides a flexible environment for developing and testing deep learning models.

Although the system shows promising results, it is primarily designed for research and academic purposes. Further improvements such as larger datasets, improved feature extraction methods, and real-time deployment can enhance the system's reliability and accuracy. Overall, this project proves that biometric analysis combined with artificial intelligence has significant potential for developing automated and intelligent healthcare prediction systems in the future.

10.2 FUTURE ENHANCEMENT

Although the Fingerprint-Based Blood Group Detection system demonstrates promising results using deep learning techniques, several improvements can be made in the future to enhance its performance, accuracy, and real-world usability. The current system mainly focuses on training deep learning models using fingerprint images; however, further development can make the system more reliable and suitable for practical healthcare applications.

One possible enhancement is the use of larger and more diverse fingerprint datasets. Increasing the dataset size with images collected from different individuals, age groups, and environments can help the deep learning models learn more complex fingerprint patterns. A larger dataset would reduce overfitting and improve the overall prediction accuracy of the system. Another improvement can be the integration of more advanced deep learning architectures such as EfficientNet, DenseNet, or Vision Transformers. These modern models have shown better performance in image classification tasks and could improve feature extraction from fingerprint images. Additionally, techniques such as data augmentation, hyperparameter tuning, and transfer learning can further enhance the model's prediction capability.

The system can also be improved by developing a real-time application or web-based interface where users can upload fingerprint images and instantly receive blood group predictions. Integrating the trained model with web frameworks or mobile applications can make the system more accessible for research demonstrations and healthcare assistance tools.

Future work may also focus on improving system security and data privacy, especially because fingerprint images are sensitive biometric data. Implementing secure data storage, authentication mechanisms, and encrypted data transmission can ensure safe usage of biometric information.

Finally, combining fingerprint analysis with other biometric or medical data sources may further improve prediction reliability. Multi-modal biometric systems that analyze fingerprints along with palm prints, iris patterns, or medical records could provide more accurate and dependable blood group prediction systems for healthcare applications.

APPENDIX A - SOURCE CODE

```
import os
import glob
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd
from sklearn.model_selection import train_test_split
from tensorflow.keras.preprocessing.image import ImageDataGenerator
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Dropout, Conv2D,
MaxPooling2D, Flatten
from tensorflow.keras.applications import ResNet50
from tensorflow.keras.applications.resnet50 import preprocess_input
from sklearn.metrics import classification_report
file_path = 'dataset_blood_group'
# List all classes in the dataset
name_class = os.listdir(file_path)
print(name_class)
# Get file paths for all images in the dataset
filepaths = list(glob.glob(file_path + '/**/*.*'))
# Example of printing a subset of file paths
print(filepaths[-6:-1])
# Extract labels from file paths
labels = list(map(lambda x: os.path.split(os.path.split(x)[0])[1], filepaths))
print(labels)
# Create a DataFrame with file paths and labels
```

```
filepath = pd.Series(filepaths, name='Filepath').astype(str)
Labels = pd.Series(labels, name='Label')
data = pd.concat([filepath, Labels], axis=1)
data = data.sample(frac=1).reset_index(drop=True)
print(data.head(5))
# Visualize class distribution
counts = data.Label.value_counts()
sns.barplot(x=counts.index, y=counts)
plt.xlabel('Type') plt.xticks(rotation=90)
plt.show()
# Split data into training and testing sets
train, test = train_test_split(data, test_size=0.20, random_state=42)
# Visualize some images from the dataset
fig, axes = plt.subplots(nrows=5, ncols=3, figsize=(10, 8),
subplot_kw={'xticks': [], 'yticks': []})
for i, ax in enumerate(axes.flat):
    ax.imshow(plt.imread(data.Filepath[i]))
    ax.set_title(data.Label[i])
plt.tight_layout()
plt.show()
# Set up ImageDataGenerator for training and validation data
train_datagen =
ImageDataGenerator(preprocessing_function=preprocess_input)
test_datagen =
ImageDataGenerator(preprocessing_function=preprocess_input)
train_gen = train_datagen.flow_from_dataframe(
    dataframe=train,
```

```
    x_col='Filepath',
    y_col='Label',
    target_size=(224, 224),  # Adjusted to match ResNet50 input size
    class_mode='categorical',
    batch_size=32,
    shuffle=True,
    seed=42
)
valid_gen =
    test_datagen.flow_from_dataframe( dataframe=test,
    x_col='Filepath',
    y_col='Label',
    target_size=(224, 224),  # Adjusted to match ResNet50 input size
    class_mode='categorical',
    batch_size=32,
    shuffle=False,
    seed=42
)
# Define the LeNet model
model = Sequential([
    Conv2D(6,  kernel_size=(5,  5),  activation='relu',  input_shape=(224,
224, 3)),
    MaxPooling2D(pool_size=(2, 2)),
    Conv2D(16, kernel_size=(5, 5), activation='relu'),
    MaxPooling2D(pool_size=(2, 2)),
    Flatten(),
    Dense(120, activation='relu'),
    Dense(84, activation='relu'),
```

```
    Dense(8, activation='softmax')
])
model.compile( optimizer="adam
    ",
    loss='categorical_crossentropy',
    metrics=['accuracy']
)
# Train the model
history = model.fit(
    train_gen,
    validation_data=valid_gen,
    epochs=20
)
# Plot training history: accuracy
pd.DataFrame(history.history)[['accuracy', 'val_accuracy']].plot()
plt.title("Accuracy")
plt.show()

# Plot training history: loss
pd.DataFrame(history.history)[['loss', 'val_loss']].plot()
plt.title("Loss")
plt.show()
# Evaluate the model on test data
results = model.evaluate(valid_gen, verbose=0)
print(f"Test Loss: {results[0]:.5f}")
print(f"Test Accuracy: {results[1]*100:.2f}%")
# Predict labels for test data
pred = model.predict(valid_gen)
```

```
pred = np.argmax(pred, axis=1)
# Map predicted labels
labels = train_gen.class_indices
labels = dict((v, k) for k, v in labels.items())
pred = [labels[k] for k in pred]
# Compare predicted labels with true labels and print classification report
y_test = list(test.Label)
print(classification_report(y_test, pred))
model.save("model_blood_group_detection_lenet.keras")
import numpy as np
import matplotlib.pyplot as plt
from tensorflow.keras.models import load_model
from tensorflow.keras.preprocessing import image
from tensorflow.keras.applications.imagenet_utils import preprocess_input
# Load the pre-trained model
model = load_model('model_blood_group_detection_lenet.keras')
# Define the class labels
labels = {'A+': 0, 'A-': 1, 'AB+': 2, 'AB-': 3, 'B+': 4, 'B-': 5, 'O+': 6, 'O-': 7}
labels = dict((v, k) for k, v in labels.items())
# Example of loading a single image and making a prediction
img_path = 'dataset_blood_group/A+/cluster_0_1001.BMP'
# Preprocess the image accordingly (check the model's expected input dimensions)
img = image.load_img(img_path, target_size=(224, 224))  # Example target size for AlexNet (224x224)
x = image.img_to_array(img)
x = np.expand_dims(x, axis=0)
```

```
x = preprocess_input(x)  # Ensure this matches the model's preprocessing
function
# Make prediction
result = model.predict(x)
predicted_class = np.argmax(result)  # Get the predicted class index
# Map the predicted class to the label
predicted_label = labels[predicted_class]
confidence = result[0][predicted_class] * 100  # Confidence level
# Display the image
plt.imshow(image.array_to_img(image.img_to_array(img) / 255.0))
plt.axis('off')  # Hide axes
# Display the prediction and confidence below the image
plt.title(f"Prediction: {predicted_label} with confidence
{confidence:.2f}%")
plt.show()
```

APPENDIX B - SCREENSHOTS

```
PS C:\Fingerprint_Project> python predict.py
Loading Model...
Model Loaded Successfully

Fingerprint Blood Group Detection System
1 - Select From Dataset
2 - Select From Storage
3 - Capture From Camera
Enter choice : 1
Detected Blood Group: O+
Result saved in results folder
```

Figure B.1 Blood Group Detection

```
PS C:\Fingerprint_Project> python predict.py
Loading Model...
Model Loaded Successfully

Fingerprint Blood Group Detection
1 - Select From Dataset
2 - Select From Storage
3 - Capture From Camera
Enter choice : 2
Predicted Output : B+

Result saved in results folder
PS C:\Fingerprint_Project>
```

Figure B.2 Blood Group Detection

```
PS C:\Fingerprint_Project> python predict.py
Loading Model...
Model Loaded Successfully

Fingerprint Blood Group Detection
1 - Select From Dataset
2 - Select From Storage
3 - Capture From Camera
Enter choice : 3
Press C to capture fingerprint
Predicted Output : O-

Result saved in results folder
PS C:\Fingerprint_Project>
```

Figure B.3 Blood Group Detection

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